Advances in Generative Modeling

Flow and Diffusion Models

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Instructions

- Choose **any four** theory problems from **Problems 1–14** and solve them; each is worth 10 points ($4 \times 10 = 40$).
- Complete **any two** experimental assignments from **Problems 15–17**; each is worth 30 points ($2 \times 30 = 60$). **Note:** Problem 17 requires access to an `A100` GPU. If you wish to apply for free `Colab Pro` access, please refer to the following post: `https://edstem.org/us/courses/81892/discussion/7240162`
- Total: 100 points. Show clear reasoning and justify all steps; include figures/logs for experimental parts.

Problem 1 (Straight paths to a single data point). **[10 pts]** Let $X_0 \sim P_0$ and suppose $P_{\text{data}} = \delta_{x^\star}$ is a single point mass at $x^\star \in \mathbb{R}^d$. Consider the straight interpolation

$$X_t = t\, x^\star + (1-t)X_0 \qquad \forall t \in [0,1].$$

1. Derive the time–dependent velocity field $v_t^*(x)$ such that the ODE $\dot{X}_t = v_t^*(X_t)$ has solution $X_t = t\, x^\star + (1-t)X_0$.
2. Show that an *arbitrary* time partition $0 = t_0 < t_1 < \cdots < t_N = 1$ with Euler updates

$$X_{t_{k+1}} \leftarrow X_{t_k} + (t_{k+1} - t_k)\, v^*_{t_k}(X_{t_k})$$

recovers exactly the straight trajectory $X_{t_k} = t_k\, x^\star + (1-t_k)X_0$ for all k, meaning that Euler updates simulate the ODE with perfect precision in this case. Explain why this is the case intuitively.

Problem 2 (Intersections and the need for an expected velocity). **[10 pts]** Let $X_t = tX_1 + (1-t)X_0$ with $X_0 \sim P_0$ and $X_1 \sim P_{\text{data}}$, drawn independently.

1. Give a concrete example (in $\mathbb{R}$ or $\mathbb{R}^2$) of two pairs (X_0, X_1) and (X_0', X_1') whose straight lines intersect at the same (x, t).
2. Explain why a deterministic ODE $\dot{Z}_t = v_t(Z_t)$ cannot assign two different velocities at the same state-time pair (x, t).

3. Explain how the velocity field in rectified flow

$$v_t^*(x) = \mathbb{E}[X_1 - X_0 \mid X_t = x]$$

can be viewed as a way to resolve intersections by "averaging directions" at collisions.

Problem 3 (Least Squares ⇒ Conditional Expectation). **[10 pts]** Let (U, V) be any pair of random variables with joint density $p(u, v)$. Consider the least-squares optimization problem:

$$\min_f \ \mathbb{E}\Big[\big\|V - f(U)\big\|^2\Big], \tag{1}$$

where the minimization is over all measurable functions f. We seek a function f that approximates V as accurately as possible using U.

1. Show that **(Bias–variance decomposition)**

$$\mathbb{E}\big[\|V - f(U)\|^2\big] = \underbrace{\mathbb{E}\big[\|V - \mathbb{E}[V \mid U]\|^2\big]}_{\text{conditional variance}} + \underbrace{\mathbb{E}\big[\|\mathbb{E}[V \mid U] - f(U)\|^2\big]}_{\text{bias}}.$$

Here $\mathbb{E}[V \mid U]$ denotes the conditional expectation of V given U, defined as

$$\mathbb{E}[V \mid U = u] \ = \ \int v\, p(v \mid u)\, \mathrm{d}v.$$

Explain the intuitive meaning of the two terms in this decomposition.

2. Prove that the minimizer of (1) is the conditional expectation:

$$f^*(u) \ = \ \mathbb{E}[V \mid U = u].$$

3. Take $U = (X_t, t)$ and $V = X_1 - X_0$. Show that

$$v_t^*(x) \ = \ \mathbb{E}[X_1 - X_0 \mid X_t = x]$$

minimizes the rectified-flow objective

$$\int_0^1 \mathbb{E}\Big[\big\|X_1 - X_0 - v_t(X_t)\big\|^2\Big]\, \mathrm{d}t.$$

Problem 4 (Two equivalent targets). **[10 pts]** Let (X_0, X_1) be any random variable pair, and let $X_t = tX_1 + (1-t)X_0$ be the straight interpolation at time t. Recall that the RF velocity equals

$$v_t^*(x) = \mathbb{E}[X_1 - X_0 \mid X_t = x].$$

Define $\mu_t^1(x) = \mathbb{E}\left[X_1 \mid X_t = x\right]$. Prove that

$$v_t^*(x) = \frac{\mu_t^1(x) - x}{1 - t}.$$

Problem 5 (Continuity Equation from a Test Function Argument). **[10 pts]** Let $\{X_t : t \in [0,1]\}$ be a stochastic process with smooth trajectories. Let p_t denote the density of X_t at each t, and assume $p_t(x)$ is smooth and positive everywhere. In this case, we define the expected velocity field of $\{X_t\}$ as

$$v_t^*(x) = \mathbb{E}[\dot{X}_t \mid X_t = x],$$

where $\dot{X}_t$ denotes the time derivative of X_t. Let $h : \mathbb{R}^d \to \mathbb{R}$ be a smooth, compactly supported test function.

1. Show that
$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[h(X_t)] = \int h(x)\,\dot{p}_t(x)\,\mathrm{d}x.$$

2. Show that
$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[h(X_t)] = \int \nabla h(x)^\top v_t^*(x)\,p_t(x)\,\mathrm{d}x.$$

3. Recall the integration by parts formula:
$$\int \nabla p(x)^\top h(x) + p(x)\,\nabla\cdot h(x)\,\mathrm{d}x = \int \nabla\cdot\big(h(x)p(x)\big)\,\mathrm{d}x = 0,$$
where $\nabla\cdot h(x) = \sum_{i=1}^d \partial_{x_i} h_i(x)$ denotes the divergence of a vector field $h : \mathbb{R}^d \to \mathbb{R}^d$.
Use integration by parts, together with (1) and (2), to derive the continuity equation:
$$\partial_t p_t(x) = -\nabla\cdot\big(v_t^*(x)\,p_t(x)\big).$$

Problem 6 (Time Weighting in the Objective). Consider the weighted objective

$$\min_v \int_0^1 w_t\,\mathbb{E}\Big[\big\|X_1 - X_0 - v_t(X_t)\big\|^2\Big]\,\mathrm{d}t,$$

where each $w_t > 0$ is a positive weight specifying the importance of the error at time t.

1. Show that if $v_t(\cdot)$ is optimized *independently for each* t (i.e., no parameter sharing across t), the minimizer is still given by
$$v_t^*(x) = \mathbb{E}[X_1 - X_0 \mid X_t = x]$$
for all t.

2. Explain why, in practice, when using a single neural network $v^\theta(x,t)$ that *shares parameters* across t, the choice of w_t can affect the learned v^θ.

Problem 7 (Neural ODE MLE (change of variables along a flow)). **[10 pts]** Let Z_t follow $\dot{Z}_t = v^\theta(Z_t, t)$ with $Z_0 \sim p_0$. Denote by p_t^θ the density of Z_t.

1. Using the chain rule and Jacobi's formula, derive

$$\log p_t^\theta(x) = \log p_0(z_0) - \int_0^t \operatorname{tr}\big(\nabla_x v^\theta(z_\tau, \tau)\big)\,\mathrm{d}\tau,$$

   where $z_\tau, \tau \in [0, t]$ is the trajectory with $z_t = x$.

2. Specialize to $d = 1$ and $v^\theta(z, t) = a(t), z$ with a scalar function $a(t)$. Write $\log p_t^\theta(x)$ in terms of $\log p_0(z_0)$ and $\int_0^t a(\tau)\mathrm{d}\tau$.

3. Briefly explain why maximum-likelihood training for v^θ is computationally expensive compared to rectified-flow training.

Problem 8 (Gaussian $\to$ Gaussian in 1D). **[10 pts]** Assume $X_0 \sim \mathcal{N}(0, 1)$, $X_1 \sim \mathcal{N}(\mu, 1)$, independent, and $X_t = tX_1 + (1-t)X_0$.

1. Compute $\mathbb{E}[X_1 \mid X_t = x]$ and $\mathbb{E}[X_0 \mid X_t = x]$.

2. Derive the rectified-flow velocity $v^*(x, t) = \mathbb{E}[X_1 - X_0 \mid X_t = x]$ and show it has the affine form $v^*(x, t) = a(t)\,x + b(t)$ for explicit $a(t), b(t)$.

3. (Concept check) Why does this v^* transport $\mathcal{N}(0, 1)$ at $t = 0$ to $\mathcal{N}(\mu, 1)$ at $t = 1$?

Problem 9 (Two-point mixture with Gaussian noise). **[10 pts]** Let $X_0 \sim \mathcal{N}(0, I_d)$ and X_1 take values $x^\star$ or $y^\star$ with equal probability $1/2$. For $t \in [0, 1)$, $X_t = tX_1 + (1-t)X_0$.

1. Using Bayes' rule, show that

$$\mathbb{P}(X_1 = x^\star \mid X_t = x) = \frac{\exp\left(-\dfrac{|x - tx^\star|^2}{2(1-t)^2}\right)}{\exp\left(-\dfrac{|x - tx^\star|^2}{2(1-t)^2}\right) + \exp\left(-\dfrac{|x - ty^\star|^2}{2(1-t)^2}\right)}.$$

2. Derive the closed-form velocity

$$v^*(x, t) = \sum_{u \in \{x^\star, y^\star\}} \omega_u(x, t)\,\frac{u - x}{1 - t},$$

   where the weights $\omega_u(x, t)$ are the posteriors from (a).

3. Provide an intuitive interpretation of (b) in terms of "pulls" from the two endpoints.

Problem 10 (Tweedie's identity and "score from RF"). **[10 pts]** Let $X_t = tX_1 + (1-t)X_0$ with $X_0 \sim \mathcal{N}(0, I)$ and $X_1 \sim P^{\text{data}}$, independent. Denote the density of X_t by ρ_t and the rectified–flow velocity by $v^*(x,t) = \mathbb{E}[X_1 - X_0 \mid X_t = x]$.

1. Show that
$$\rho_t(x) \propto \int \rho_1(x_1) \exp\left(-\frac{|x - tx_1|^2}{2(1-t)^2}\right) \mathrm{d}x_1.$$

2. Differentiate $\log \rho_t(x)$ to prove the Tweedie identity
$$\nabla \log \rho_t(x) = \mathbb{E}\left[\frac{tX_1 - x}{(1-t)^2} \,\middle|\, X_t = x\right].$$

3. Using the interpolation identity, verify that
$$\mathbb{E}[X_1 \mid X_t = x] = x + (1-t)\, v^*(x,t),$$
and conclude the useful relation
$$\nabla \log \rho_t(x) = \frac{t\, v^*(x,t) - x}{1-t}.$$

Problem 11 (Endpoint behavior and a safe noise schedule). **[10 pts]** Using $\nabla \log \rho_t(x) = \big(t\, v^*(x,t) - x\big)/(1-t)$, the Langevin drift magnitude scales like $\sigma_t^2/(1-t)$ near $t = 1$.

1. Suppose $\sigma_t = c(1-t)^\alpha$ with $c > 0$. Find the condition on α so that $\dfrac{\sigma_t^2}{1-t}$ remains bounded as $t \to 1$.

2. Interpret your answer: what happens if $\alpha < \frac{1}{2}$, $\alpha = \frac{1}{2}$, and $\alpha > \frac{1}{2}$?

3. Briefly explain why one often lets $\sigma_t \to 0$ as $t \to 1$ in practice.

Problem 12 (Euler–Maruyama for RF+Langevin and its moments). **[10 pts]** Consider a single time step of size Δt for the hybrid SDE at state x and time t. Write the Euler–Maruyama update both in "score form"
$$Z_{t+\Delta t} = x + \Big[v^*(x,t) + \sigma_t^2 \nabla \log \rho_t(x)\Big]\Delta t + \sqrt{2}\,\sigma_t \sqrt{\Delta t}\,\xi, \quad \xi \sim \mathcal{N}(0, I),$$
and in "RF-only form" by substituting $\nabla \log \rho_t(x) = \dfrac{t\, v^*(x,t) - x}{1-t}$.

Problem 13 (Single-point data under RF+Langevin). **[10 pts]** Assume $P^{\text{data}} = \delta_{x^\star}$ and $X_0 \sim \mathcal{N}(0, I)$. Then $X_t = t\, x^\star + (1-t)X_0$.

1. Show that $v^*(x,t) = \dfrac{x^\star - x}{1-t}$ and $\nabla \log \rho_t(x) = \dfrac{t\,x^\star - x}{(1-t)^2}$.

2. Write the hybrid drift

$$b(x,t) = \frac{x^\star - x}{1-t} + \sigma_t^2 \, \frac{t\,x^\star - x}{(1-t)^2}.$$

Explain how the second term "pulls" toward $x^\star$ and how its strength depends on t and σ_t.

3. Verify that when $\sigma_t \equiv 0$ you recover the exact straight-line ODE whose one-step Euler update from $t = 0$ to $t = 1$ is exact.

Problem 14 (Discrete-time Langevin (ULA): stability and stationary bias in the Gaussian case). **[10 pts]** Consider the discrete update $x_{k+1} = x_k + \varepsilon \, \nabla \log p(x_k) + \sqrt{2\varepsilon}\, \xi_k$, $\xi_k \sim \mathcal{N}(0, I)$ i.i.d., with $p(x) \propto \exp(-\frac{1}{2}\|x - \mu\|^2)$ so that $\nabla \log p(x) = \mu - x$.

1. Show that the recursion can be written as

$$x_{k+1} = (1-\varepsilon)x_k + \varepsilon\,\mu + \sqrt{2\varepsilon}\,\xi_k.$$

2. In 1D, prove mean-square stability holds iff $|1-\varepsilon| < 1$, i.e., $0 < \varepsilon < 2$. (Hint: analyze $\mathbb{E}\big[(x_k - \mu)^2\big]$.)

3. Compute the stationary variance in 1D:

$$\mathrm{Var}_\infty = \frac{2\varepsilon}{1-(1-\varepsilon)^2} = \frac{1}{1-\varepsilon/2}.$$

Explain why $\mathrm{Var}_\infty \to 1$ as $\varepsilon \to 0$, but $\mathrm{Var}_\infty > 1$ for any fixed $\varepsilon > 0$, demonstrating a discretization bias. (Generalize coordinate-wise to $\mathbb{R}^d$.)

Problem 15 (Rectified Flow on 2D Toy Data and Real Data). **[30 pts]** In this problem, you will explore the *Rectified Flow* framework for learning deterministic transport between a source distribution P_0 and a target distribution P_1. You will start from a simple 2D toy setup.

1. **(10 pts) Run the provided 2D Toy Example.** Open the implementation from Colab. Run the code end-to-end to train the *Rectified Flow* on two Gaussian mixture models, visualize the learned trajectories.

2. **(10 pts) Apply Rectified Flow to more 2D dataset.** Use the provided `HM4` dataset. Replace the synthetic GMM data in the 2D toy example from **Homework 4 (GAN 2D Toy Dataset)**: `https://drive.google.com/file/d/1b5Zeqm8G4vXBe2ML1FjixZbE4EmR7b9O/view?usp=sharing`

   Re-run the notebook and record your observations. Experiment with different hyperparameters such as learning rate, number of epochs, and batch size to analyze their effects on training and generation quality.

3. **(10 pts) Implement the RF loss manually.** Instead of using the built-in `get_loss` function in the provided implementation, write your own version following the theoretical formulation:

$$\mathcal{L}(v_t) = \mathbb{E}_{t\sim U[0,1],\,(x_0,x_1)\sim(P_0,P_1)}\left[\|x_1 - x_0 - v_\theta(x_t,t)\|^2\right].$$

   Show that your custom implementation produces consistent training behavior with the provided one.

Problem 16 (Rectified Flow on MNIST Generation). **[30 pts]** In this problem, you will train and evaluate a *Rectified Flow* model on the MNIST dataset to learn a continuous transport from noise to handwritten digit images.

1. **(10 pts) Train a Rectified Flow model on MNIST.** Use the provided Colab notebook: Colab: MNIST Rectified Flow. Run the notebook to train the model on the MNIST dataset. If you have access to an `A100` GPU, set the number of training epochs to around 100–200 for improved results. If you do *not* have access to an `A100` GPU, your training will be relatively slower; running for about 20 epochs is acceptable and the generation quality will be moderate—that's fair for evaluation. Record the training loss curve and visualize generated samples at different training stages.

2. **(10 pts) Analyze the effect of sampling steps.** After training, vary the number of Euler integration steps (`num_steps`) used during sampling. Compare the generated image quality when using a small number of steps (e.g., 2–10) versus a large number (50-100). Discuss how the number of steps affects sample sharpness and consistency.

3. **(10 pts) Explore conditional generation and CFG.** Study the provided code for conditional generation and *Classifier-Free Guidance (CFG)*. Briefly describe how CFG modifies the velocity prediction:

$$v_t^{\mathrm{CFG}}(x) = v_t(x,\varnothing) + s\big(v_t(x,y) - v_t(x,\varnothing)\big),$$

where s is the CFG scale. You may conduct your own research online. Run experiments with different CFG scales (e.g., $s \in \{0.5, 1.0, 2.0, 3.0\}$) and record how the guidance strength affects generation quality and diversity.

Problem 17 (Text-to-Image Inference & Editing with FLUX Rectified Flow). **[30 pts]** In this assignment you will play with a **state-of-the-art** rectified-flow text-to-image model, **FLUX.1-dev**, which uses a large DiT backbone ("$\sim$13B"-scale diffusion transformer) and produces high-fidelity 1024×1024 images with strong prompt adherence. You will (i) run pure inference with several RF samplers and (ii) perform image editing with several methods. Because this model is very large, please select an A100 GPU **(80GB High GPU VRAM version)** in Colab to ensure successful execution.

- Image Generation notebook: Colab
- Image Editing notebook: Colab

1. **(15 pts) FLUX Inference with Rectified Flow.** In this exercise, you will explore inference behavior of the provided `FluxPipeline` wrapped by `FluxWrapper` and `RectifiedFlow`.
   (a) Perform inference to generate images for three different text prompts.
   (b) Compare two or more Rectified Flow samplers. Vary the time discretization num_steps and analyze its effects on image sharpness, consistency, and potential artifacts.
   (c) Sweep the *Classifier-Free Guidance* (CFG) scale s (e.g., $s \in 1.0, 2.0, 3.5, 7.5, 10$) and discuss the observed quality–diversity trade-off. Use the standard CFG formulation:

$$v_t^{\text{CFG}}(x) = v_t(x, \varnothing) + s\left(v_t(x, y) - v_t(x, \varnothing)\right).$$

2. **(15 pts) Image Editing with FLUX**
   (a) Run the provided notebook and reproduce the given image editing example using the FLUX pipeline and Rectified Flow. Show the final edited output.
   (b) Replace the image with one of your choice and experiment with editing parameters (e.g., noise level, `start_t`, `end_t`, `eta_base`, schedule type). Explore different prompts and samplers to create your own editing results.